



LOW RISK

GRADE A · PROFESSIONAL

BitGuard Token & Liquidity Locker

LOCK · Cronos · Token / liquidity locker

0xd1d20656a72048f9b19a97fc03ec9eeef6d51270

Our Auditors reviewed the Cronos locker contract 0xd1d20656a72048f9b19a97fc03ec9eeef6d51270, identified as BitGuard Token & Liquidity Locker (LOCK), and discovered the information below. A Two-Step process was used to analyze the smart contract source code; first using AI-powered security audit tools and then manually. A high-severity reentrancy surface remains where external value calls can occur relative to state updates, even though a gate modifier is present in source. Coverage is incomplete: independent scans did not return a full report, so the rating rests on chain metadata and heuristics only. Lock and vesting paths depend on block timestamp and number, which producers can bias within protocol bounds and must not be treated as randomness. Lock-level Ownable administration is still single-key unless each lock owner renounces or wraps control. The address matches a documented BitGuard V1 locker core (immutable deployment, two-step lock ownership, forward-only extend, no admin drain of customer locks); confirm the live explorer page against that seal.

LIQUIDITY	VERIFIED	BUY / SELL TAX	LP LOCKED
—	Yes	— / —	—

Identity

FIELD	VALUE
Name / symbol	BitGuard Token & Liquidity Locker / LOCK
Address	0xd1d20656a72048f9b19a97fc03ec9eeef6d51270
Network	Cronos
Standard	Token / liquidity locker
Decimals	—
Supply	—
Creator	—
Creation tx	—
Compiler	—
Optimization	—
Proxy	No
Implementation	—
Program ID	—
Bytecode size	10538 bytes
Issued	Wed, 02 Sep 2026 23:54:30 GMT

Control matrix

STATUS	CONTROL	DETAIL
IDENTITY	—	—

STATUS	CONTROL	DETAIL
INFO	Contract class	Classified as Token / liquidity locker · known deployment "BitGuard Token & Liquidity Locker".
TOKENOMICS	—	—
INFO	Liquidity	Not applicable. This is not a tradable token, so DEX pool quotes are out of scope.
MARKET	—	—
INFO	Listed on a DEX	Not applicable. DEX listings apply to tokens, not this contract class.
OWNERSHIP	—	—
PASS	Locker upgradeability	No EIP-1967 implementation slot observed.
PASS	Admin drain / emergency withdraw	BitGuard V1 documents no emergencyWithdraw on customer assets.
PASS	Unlock can be shortened	BitGuard V1 is forward-only: newUnlockTime must be "e current
IDENTITY	—	—
PASS	Source verification	Verified source is published.
UNKNOWN	Compiler / version	Compiler metadata not returned.
PASS	Proxy / implementation	No EIP-1967 implementation slot set.
PASS	Upgradeability	No upgrade entrypoint observed.
PASS	Token standard	ERC-20 / BEP-20 surface (name, symbol, decimals, totalSupply).
OWNERSHIP	—	—
UNKNOWN	Owner privileges	Owner not readable.
PASS	Hidden / admin roles	No hidden-owner flag.
PASS	Mint authority	Not flagged mintable.
PASS	Pause authority	No pause switch observed.
PASS	Blacklist / whitelist	No blacklist or whitelist gate observed.
PASS	Ownership can be restored	No take-back flag.
PASS	Owner can change balances	No arbitrary balance-write flag.
PASS	Upgrade authority	No proxy admin.
UNKNOWN	Owner is a contract (possible multisig)	Owner contract check unavailable.
PASS	Self-destruct	No self-destruct observed.
BYTECODE PRIVILEGES	—	—
PASS	Mint function in bytecode	No mint selector in bytecode.
PASS	Pause function in bytecode	No pause selector in bytecode.
PASS	Blacklist function in bytecode	No blacklist selector in bytecode.
PASS	Upgrade function in bytecode	No upgrade selector in bytecode.
PASS	Trading switch in bytecode	No trading-switch selector in bytecode.
PASS	Fee-wallet setter in bytecode	No fee-wallet setter selector.
INFO	Privileged entrypoints	2 known privileged selectors in bytecode.
TOKENOMICS	—	—
UNKNOWN	Current supply	totalSupply() not readable.
INFO	Maximum supply	Max supply not exposed.

STATUS	CONTROL	DETAIL
PASS	Mintability	Mint disabled or not present.
UNKNOWN	Largest wallet	Holder table unavailable.
UNKNOWN	Top-10 concentration	Holder table unavailable.
UNKNOWN	Owner share	Owner share not reported.
UNKNOWN	Creator share	Creator share not reported.
UNKNOWN	Holder count	Holder count not reported.
MALICIOUS BEHAVIOR	—	—
UNKNOWN	Honeypot behavior	No honeypot flag.
UNKNOWN	Buy/sell simulation	No trade simulation.
PASS	Buy restriction	Buys not flagged blocked.
PASS	Sell-all restriction	No sell-all trap flag.
UNKNOWN	Transfer tax	Buy —% / sell —%.
PASS	Dynamic / hidden tax	No owner tax-setter observed.
PASS	Per-wallet tax	No per-wallet tax flag.
PASS	Trading enable/disable	No trading enable switch observed.
PASS	Suspicious external calls	No external-call flag.
PASS	Creator honeypot history	No honeypot-history flag on the creator.
PASS	Fake-token flag	Not flagged as a fake token.
PASS	Trading cooldown	No trading cooldown flag.
PASS	Anti-whale / max-wallet	No anti-whale cap observed.
PASS	Anti-whale is mutable	Anti-whale settings are not flagged mutable.
PASS	Gas-abuse pattern	No gas-abuse flag.
PASS	Transfers pausable	Transfers not flagged pausable.

Findings

SEV	TITLE	DETAIL
HIGH	Value call pattern (reentrancy surface)	External value calls before state updates are the classic ERC-20 / ETH reentrancy pattern. · Remediation: Checks-effects-interactions plus a nonReentrant modifier on all value-moving functions. · SC05 Reentrancy
MEDIUM	Limited scanner coverage	Independent security scans did not return a full report. Score is based on chain data and heuristics only. · Remediation: Remediate on-chain and re-run this assessment before treating the finding as closed.
LOW	Block timestamp / number used in logic	Miners and sequencers can bias timestamps within bounds; not a source of randomness. · Remediation: Do not use block values for lottery, mint rarity, or fee lottery logic. · SC09 Insecure Randomness
INFO	Locker / vesting primitives in source	This looks like a lock or vesting vault rather than a tradable token. · Remediation: Confirm unlock cannot be pulled forward and that no admin drain exists. · SC01 Access Control
INFO	Ownable pattern present	A single owner key can administer the token unless renounced or wrapped in a multisig. · Remediation: Renounce or migrate owner to a documented multisig with a timelock. · SC01 Access Control

SEV	TITLE	DETAIL
INFO	Known BitGuard Token & Liquidity Locker	This address matches a documented BitGuard V1 locker core (immutable, two-step ownership, forward-only extend, no admin drain of customer locks). Still verify the live explorer page against this seal. · Remediation: Keep lock records (ID, vault, owner wallet, unlock UTC) with the client file.

OWASP Smart Contract Top 10

Each row is the risk in plain language, and whether this contract showed it.

STATUS	ID	WHAT THIS MEANS	THIS TOKEN
INFO	SC01	Who is allowed to mint, pause, upgrade, or freeze — and can they do it alone?	Found in this contract: Locker / vesting primitives in source; Ownable pattern present.
PASS	SC02	Does the contract trust a spot price that a flash loan could move?	Nothing of this type showed up in the verified source.
PASS	SC03	Does the business logic do something unexpected (wrong tax, wrong cap, hidden branch)?	Nothing of this type showed up in the verified source.
PASS	SC04	Are addresses, amounts, and caller roles checked before anything is changed?	Nothing of this type showed up in the verified source.
FAIL	SC05	Can an external call come back in and drain the contract before it finishes?	Found in this contract: Value call pattern (reentrancy surface).
PASS	SC06	Are low-level calls checked, or can they fail silently?	Nothing of this type showed up in the verified source.
PASS	SC07	Can borrowed liquidity in the same transaction trick the contract?	Nothing of this type showed up in the verified source.
PASS	SC08	Can numbers wrap around (old compilers, unchecked math) and mint extra tokens?	Nothing of this type showed up in the verified source.
WARN	SC09	Is “randomness” taken from the block, which a validator can bias?	Found in this contract: Block timestamp / number used in logic.
PASS	SC10	Can one user jam the contract so nobody else can move funds?	Nothing of this type showed up in the verified source.

Summary

On Cronos, address 0xd1d20656a72048f9b19a97fc03ec9eeef6d51270 is a verified BitGuard Token & Liquidity Locker (LOCK) contract, presented as a token/liquidity locker rather than a circulating ERC-20. Explorer verification is present; the deployment is not a proxy and is not upgradeable. Bytecode size is consistent with a self-contained locker core. Source describes an immutable V1 vault: locks and vesting schedules hold customer assets with no admin withdrawal path for those balances, and the deployer address is informational only.

Authority is split. Protocol fees route through an immutable fee-controller reference set at construction. Each lock has its own owner and pendingOwner with a two-step transfer and renounce path; administration of a lock is therefore a single EOA unless the holder moves it to a multisig or renounces. Kind-gated create paths support simple locks, LP-style locks, and vesting, with unlock extension recorded as forward-only history.

Residual risk is concentrated in interaction order (value call / reentrancy surface), timestamp-driven unlock and vesting math, and the absence of a complete independent scan corpus. Heuristic score 87 (grade A, low verdict) does not substitute for full adversarial review of live fee-controller behavior, token edge cases (fee-on-transfer, non-standard ERC-20), and operator key hygiene on individual locks.

Recommendations

#	ACTION
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#	ACTION
1	Keep every external token or ETH transfer strictly after state finalization and inside the existing nonReentrant gate; eliminate any path that sends value then mutates lock or vesting fields.
2	Commission a full manual and differential review of the deployed bytecode and fee-controller integration; do not treat heuristic-only coverage as complete.
3	Document and bound all block.timestamp / block.number uses in unlock and vesting math; never rely on them as entropy.
4	Require multisig or renounce for high-value lock ownership; treat single-EOA lock owners as a residual admin risk.
5	Re-verify the live explorer source and security metadata against the documented V1 locker seal before any production reliance.
6	Threat-model the immutable fee controller: feeFor and feeRecipient changes (if any upstream) must not enable griefing or unexpected lock creation costs.
7	Add explicit handling and tests for fee-on-transfer and non-standard ERC-20 return data so locked amounts always equal balances actually received.
8	Publish operator runbooks for extend, claim, and rescue paths so users can confirm no customer-lock drain surface exists on this deployment.

Scope & methodology

SCOPE
Contract / program identity and deployment provenance
Ownership, roles, mint / freeze / pause / upgrade authorities
Tokenomics, supply, holders, liquidity concentration
Malicious-token behavior (honeypot, tax, blacklist, traps)
EVM vulnerability classes aligned to OWASP Smart Contract Top 10
Solana SPL / Token-2022 extensions and custom program risk
METHODOLOGY
On-chain RPC reads of bytecode, storage slots (EIP-1967), ERC-20 / SPL metadata, and Token-2022 TLV extensions.
Verified-source retrieval (Sourcify / Blockscout) followed by a dedicated static-analysis engine — not an LLM — covering OWASP Smart Contract Top 10 classes and token-trap patterns.
Selector extraction from deployed bytecode to detect privileged entypoints even when source is unverified.
Independent market and liquidity quotes from multiple public feeds; missing quotes are reported as coverage gaps, not zero liquidity.
Security-signal aggregation (honeypot simulation, mint/freeze authorities, holder concentration) with vendor names withheld from the sealed letter.
BitGuard synthesis of the already-computed evidence into an engagement letter. The model does not invent on-chain facts.

Disclaimer

BitGuard Professional Smart Contract Assessment. Automated static analysis, on-chain reads, authority mapping, and multi-model review of publicly available data. This engagement snapshot is not a substitute for a human formal-verification review, legal opinion, or insurance underwriting. Privileged keys, upgrades, and off-chain operators can change after this report is sealed.